POPULATION AND SYSTEMS ECOLOGY

Module map

- Module 15: Population and Systems Ecology
- Connected lab: lab-15_population-systems-ecology.md
- Use this deck as the visual path through the module's evidence and retrieval work.

Population growth models

Limits and carrying capacity

Matter cycling and systems feedback

Module 15

Community interactions

Energy flow through trophic levels

lab-15_population-systems-ecology.md

Teaching note: Open by naming the module claim and pointing students to the lab as the evidence surface.

POPULATION AND SYSTEMS ECOLOGY

Learning objectives

- Compare exponential and logistic population growth.
- Explain carrying capacity and limiting factors.
- Distinguish density-dependent and density-independent regulation.
- Use trophic levels and the 10 percent rule to explain energy limits.
- Contrast energy flow with nutrient cycling.

OBJECTIVE 1
Compare exponential and logistic population growth.

OBJECTIVE 2
Explain carrying capacity and limiting factors.

OBJECTIVE 3
Distinguish density-dependent and density-independent regulation.

OBJECTIVE 4
Use trophic levels and the 10 percent rule to explain energy limits.

OBJECTIVE 5
Contrast energy flow with nutrient cycling.

Teaching note: Frame objectives as actions students should be able to perform, not facts to copy.

POPULATION AND SYSTEMS ECOLOGY

Topic sequence

- Population growth models: Population size changes through births, deaths, immigration, and emigration.
- Limits and carrying capacity: Exponential growth describes rapid increase; logistic growth includes carrying capacity.
- Community interactions: Density-dependent and density-independent factors limit populations differently.
- Energy flow through trophic levels: Energy flows through producers, consumers, and decomposers with loss at transfers.
- Matter cycling and systems feedback: Matter cycles through ecosystems while energy must continually enter.

TOPIC 1
Population growth models

TOPIC 2
Limits and carrying capacity

TOPIC 3
Community interactions

TOPIC 4
Energy flow through trophic levels

TOPIC 5
Matter cycling and systems feedback

Teaching note: Show the module as an ordered explanation so students can locate each new idea.

POPULATION AND SYSTEMS ECOLOGY

Module 15: Population and Systems Ecology Evidence Map

- Module focus: Population and Systems Ecology
- Central claim: Population and Systems Ecology explains population systems governed by growth, limits, interactions, energy, and feedback.
- Key nodes to connect: Population growth models, Limits and carrying capacity, Lab evidence
- Reasoning move: use 'requires' to explain relationships, not memorize terms.
- Lab connection: lab-15_population-systems-ecology.md supplies an evidence surface for the map.

Teaching note: Ask students to explain one edge in their own words before moving on.

POPULATION AND SYSTEMS ECOLOGY

Module 15: Population and Systems Ecology Reasoning Sequence

- Module focus: Population and Systems Ecology
- Trace the model: Population growth models -> Limits and carrying capacity -> Community interactions -> Energy flow through trophic levels
- Inputs become outputs: Population growth models, Limits and carrying capacity -> Compare exponential and logistic population growth., Explain carrying capacity and limiting factors.
- Feedback check: lab-15_population-systems-ecology.md evidence checks whether students can explain community interactions.

Teaching note: Emphasize where the process can fail, slow down, or be revised by evidence.

POPULATION AND SYSTEMS ECOLOGY

Terms as evidence handles

- Population: Members of one species in an area.
- Exponential growth: Growth pattern with increasing rate under abundant resources.
- Logistic growth: Growth that slows as population approaches carrying capacity.
- Carrying capacity: Approximate environment-supported population size.
- Density-dependent factor: Limiting factor whose impact changes with crowding.
- Density-independent factor: Limiting factor not directly tied to density.

POPULATION
Members of one species in an area.

EXPONENTIAL GROWTH
Growth pattern with increasing rate under abundant resources.

LOGISTIC GROWTH
Growth that slows as population approaches carrying capacity.

CARRYING CAPACITY
Approximate environment-supported population size.

DENSITY-DEPENDENT FACTOR
Limiting factor whose impact changes with crowding.

DENSITY-INDEPENDENT FACTOR
Limiting factor not directly tied to density.

Teaching note: Treat terms as handles for reasoning. Each term should help explain a claim or observation.

POPULATION AND SYSTEMS ECOLOGY

Lab connection

- Lab file: lab-15_population-systems-ecology.md
- Lab evidence should test or illustrate: Community interactions
- Students should leave with a concrete observation, comparison, or model tied to the module claim.

QUESTION
Population growth models

EVIDENCE
lab-15_population-systems-ecology.md

CLAIM
Compare exponential and logistic population growth.

Teaching note: Make the lab connection explicit before students encounter the activity.

POPULATION AND SYSTEMS ECOLOGY

Contrast check

- Do not stop at: Population size changes through births, deaths, immigration, and emigration.
- Stronger explanation: Matter cycles through ecosystems while energy must continually enter.
- The goal is to move from a named idea to a testable biological explanation.

SURFACE
Population size changes through births, deaths, immigration, and emigration.

DEEPER
Matter cycles through ecosystems while energy must continually enter.

Teaching note: Use this slide to prevent shallow vocabulary-only learning.

POPULATION AND SYSTEMS ECOLOGY

Module 15: Population and Systems Ecology Retrieval and Lab Check

- Module focus: Population and Systems Ecology
- Retrieval prompt: What changes population size?
- Answer check: Compare exponential and logistic population growth.
- Required terms: Population, Exponential growth, Logistic growth
- Lab connection: lab-15_population-systems-ecology.md; lab-15_population-systems-ecology.md supplies evidence for ecology evidence from population

Teaching note: Pause here. Students answer first without notes, then compare to the checks.

POPULATION AND SYSTEMS ECOLOGY

Practice quiz bridge

- 1. Which statement best defines Population in Module 15?
- 2. Which learning goal best supports the topic "Limits and carrying capacity"?
- 3. How should lab-15_population-systems-ecology.md support Module 15?
- 4. Which retrieval move best prepares a student for Population and Systems Ecology?

QUIZ 1
A

QUIZ 2
B

QUIZ 3
C

QUIZ 4
D

Teaching note: Use the quiz as formative feedback. Answer key: 1=A, 2=B, 3=C, 4=D

POPULATION AND SYSTEMS ECOLOGY

Synthesis and exit ticket

- Exit claim: explain population growth models using evidence from lab-15_population-systems-ecology.md.
- Must include: Population, Exponential growth, Logistic growth.
- Revision prompt: what evidence would change your explanation?

CLAIM
Population growth models

EVIDENCE
lab-15_population-systems-ecology.md

REVISION
What would change your mind?

Teaching note: End with a claim-evidence-reasoning exit ticket connected to the same module data.